

中国科学院大学
University of Chinese Academy of Sciences

Generative Models & AI4S Applications

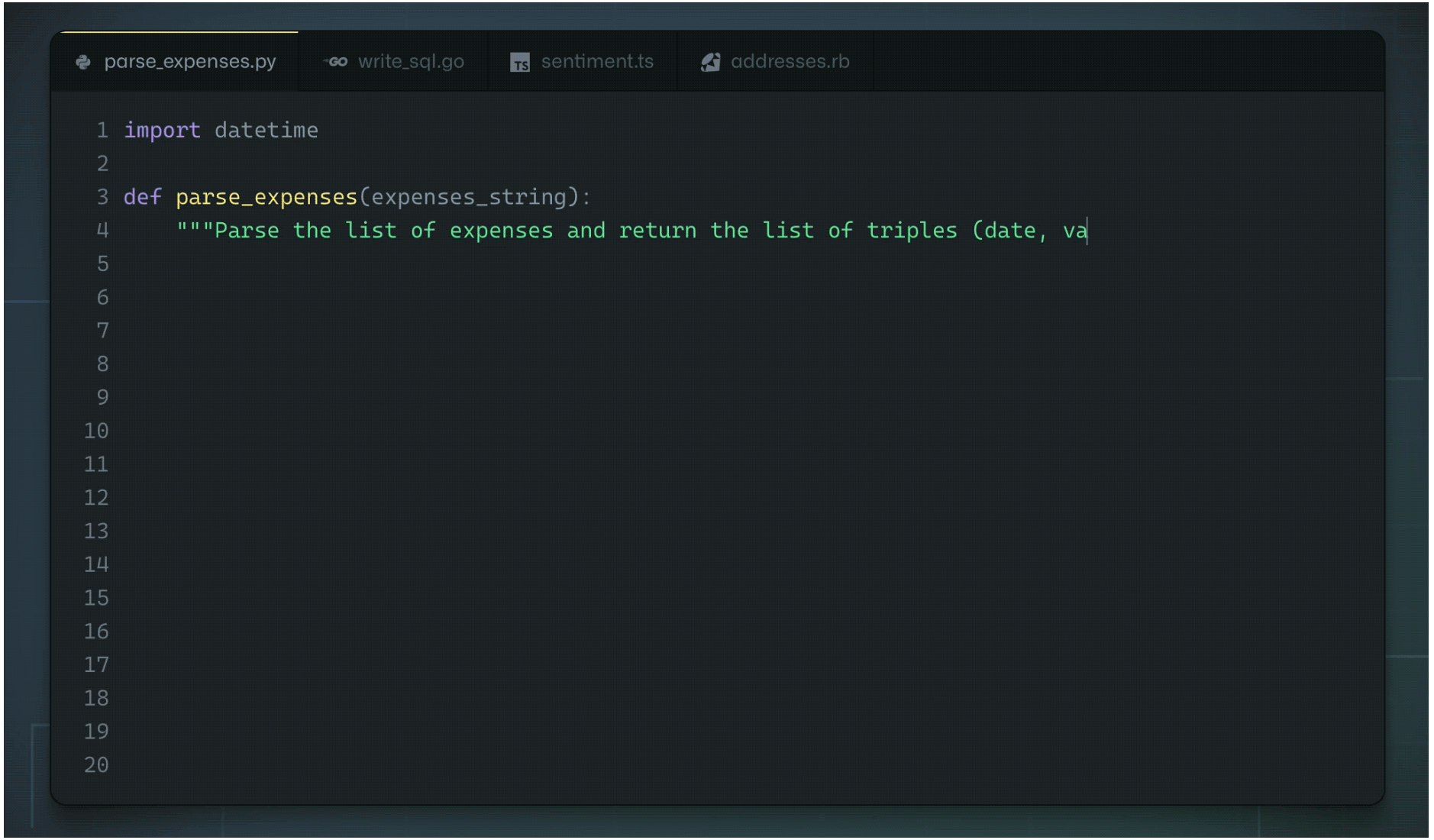
Lecture 1 Introduction

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The Era of “Generative AI”



Coding Agent (Copilot)

The Era of “Generative AI”



Prompt: teddy bear teaching a course, with "generative models" written on blackboard

Text-to-image generation: Stable Diffusion

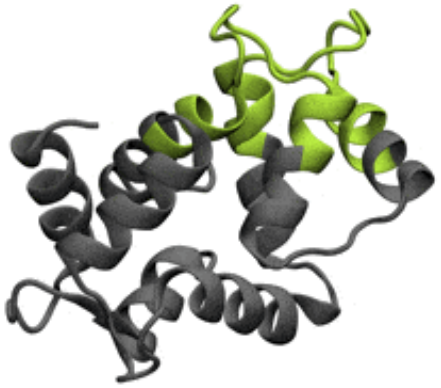
Figure adapted from MIT 6.S978

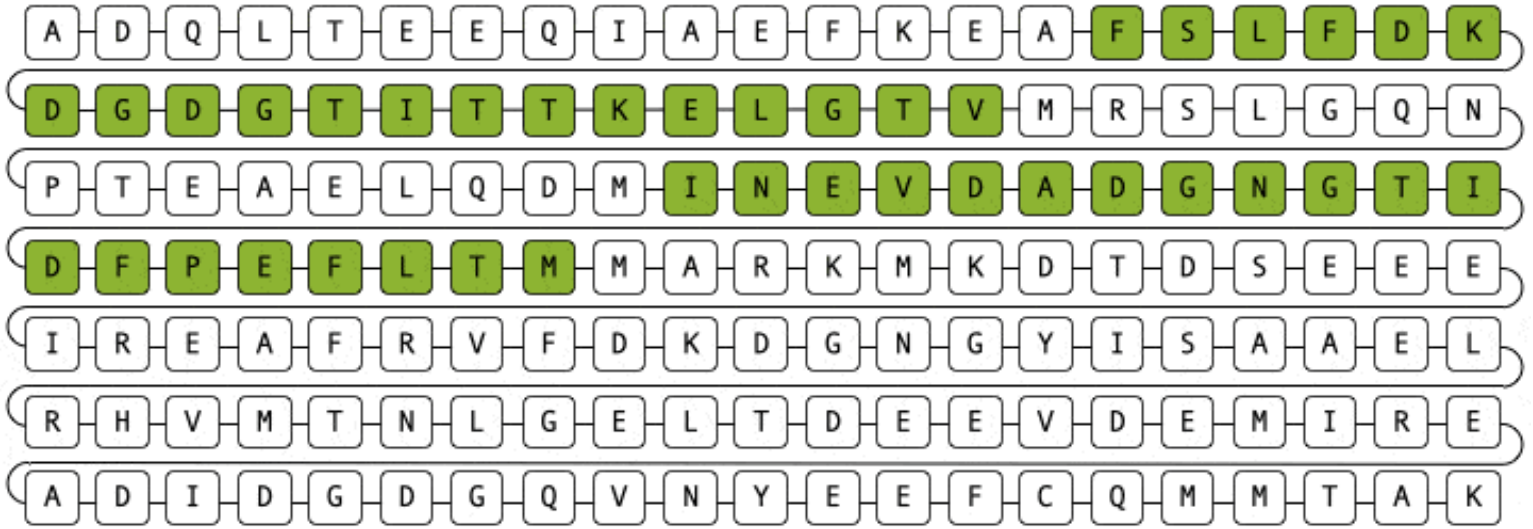
The Era of “Generative AI”



Text-to-video generation: Sora @ OpenAI

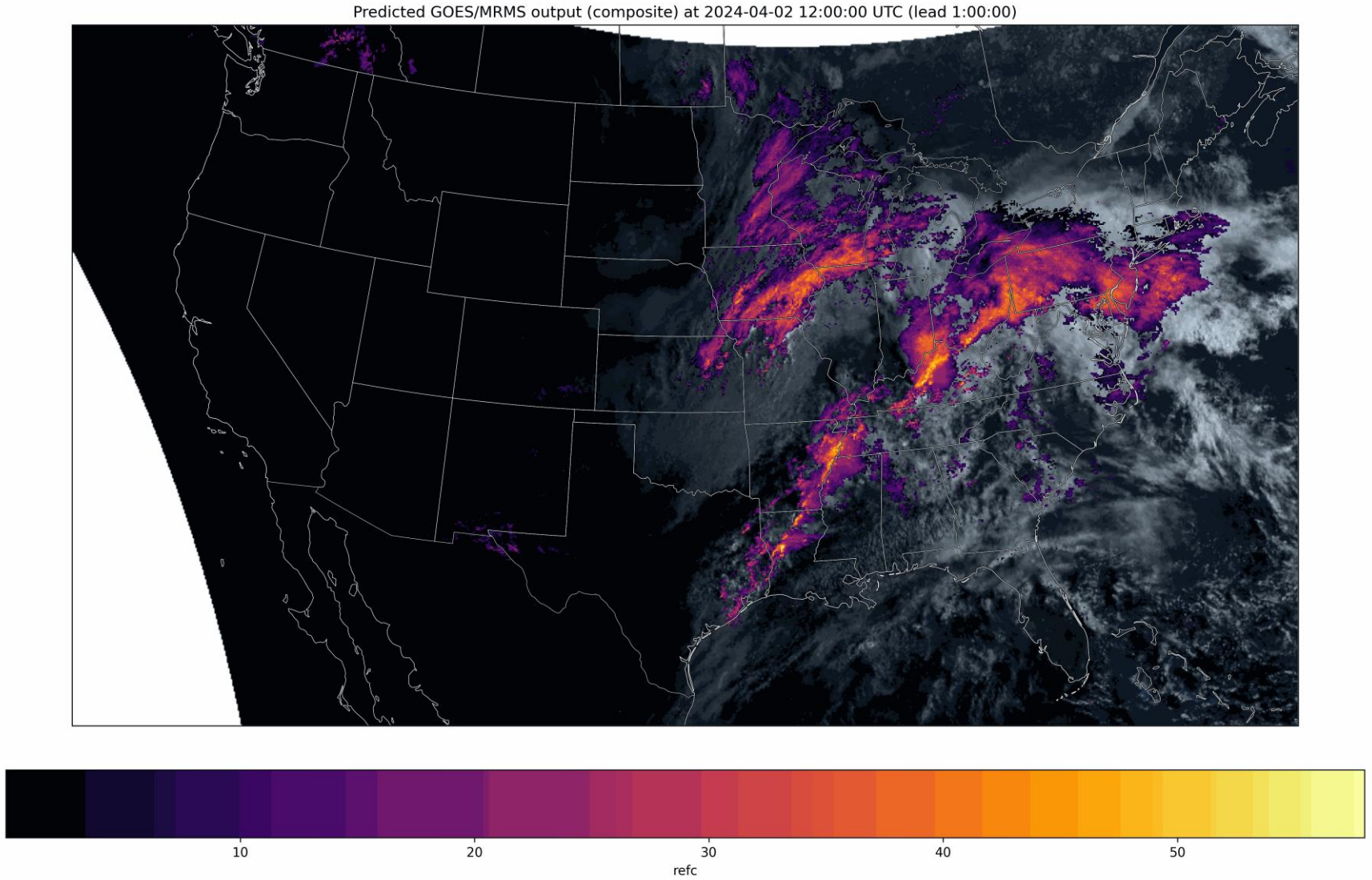
The Era of “Generative AI”





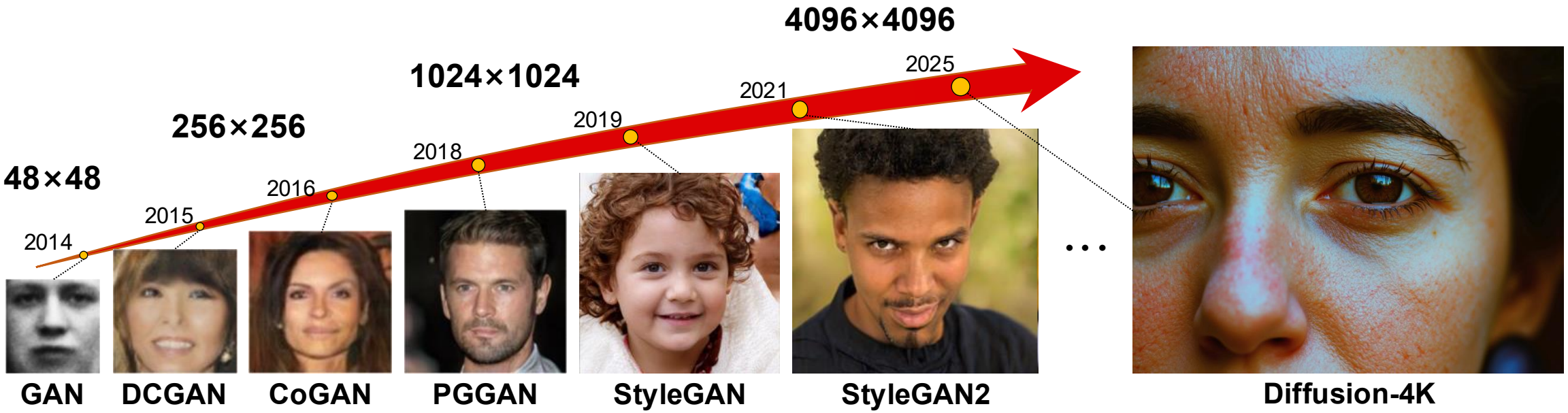
AI for Science: Protein Design

The Era of “Generative AI”



AI for Science: Weather Forecasting

Generative Models before the “Generative AI” Era

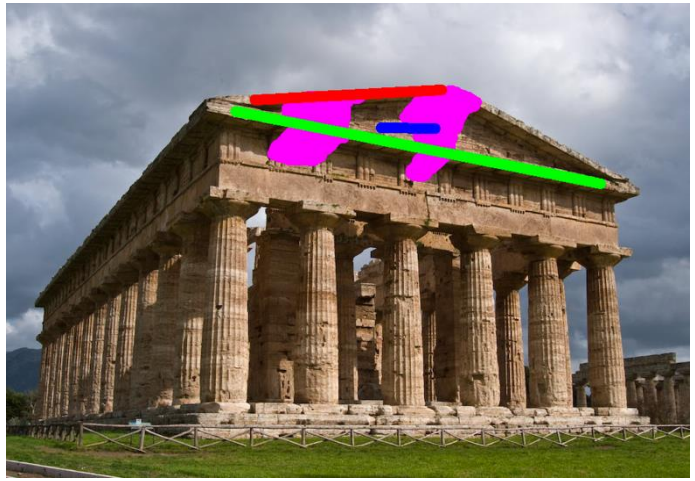


Progression of Synthetic Face Quality

Generative Models before the “Generative AI” Era



Input Image



Hole + Constraints



Hole Filled



Constraints

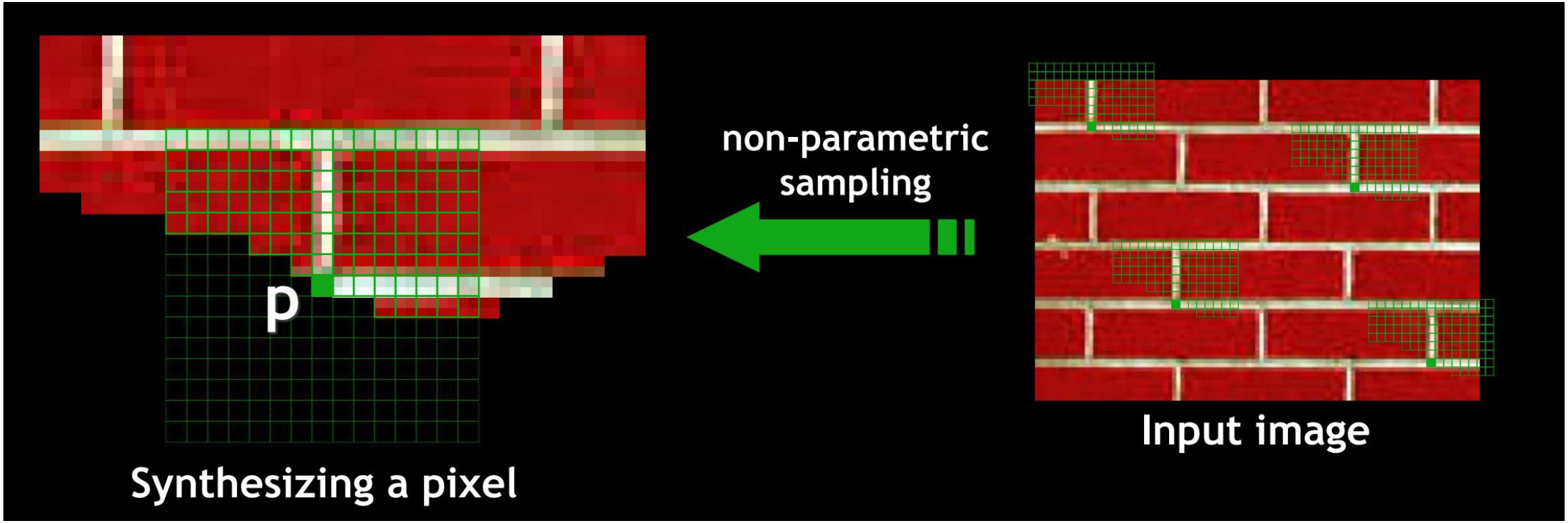


Retargeted

PatchMatch – Content-aware Fill in Photoshop (**SIGGRAPH, 2009**)

Generative Models before the “Generative AI” Era

Texture Synthesis



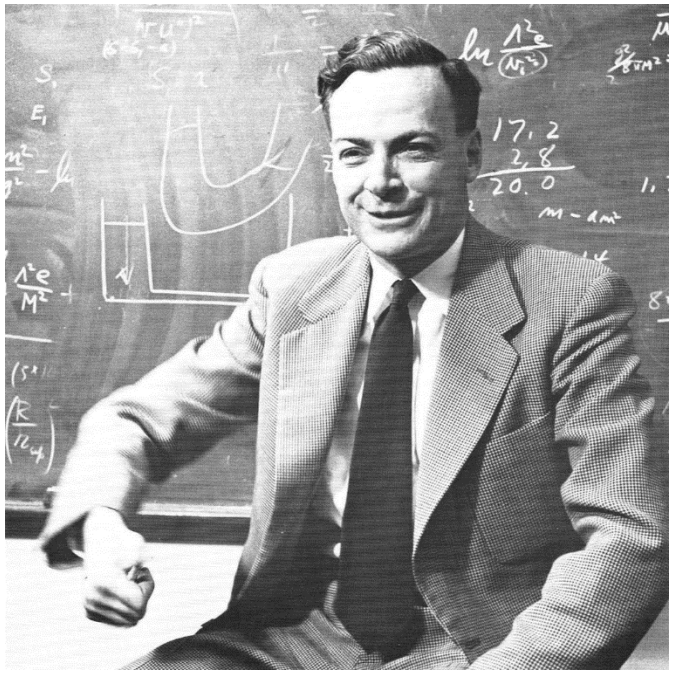
Synthesized Output Image

Reference Texture Image

Efros-Leung Algorithm (ICCV, 1999)

What are Generative Models?





Richard Feynman: "*What I cannot create, I do not understand*"

Generative modeling: "*What I understand, I can* ***create***"

A Toy Example: Scalar Generation

- **Task:** "Generate" plausible body weights x for students (e.g., male)
- **Modeling assumption:**

$$x \sim p_{data}(x) \quad \Rightarrow \quad p_\theta(x) = \mathcal{N}(x; \mu, \sigma^2) \approx p_{data}(x)$$

Unknown data distribution **Approximated by Gaussian**

  - If $\boldsymbol{\theta} = [\mu, \sigma^2]^\top$ is known, new samples x_{new} can be sampled from $p_\theta(x)$

$$x_{new} \sim p_\theta(x)$$

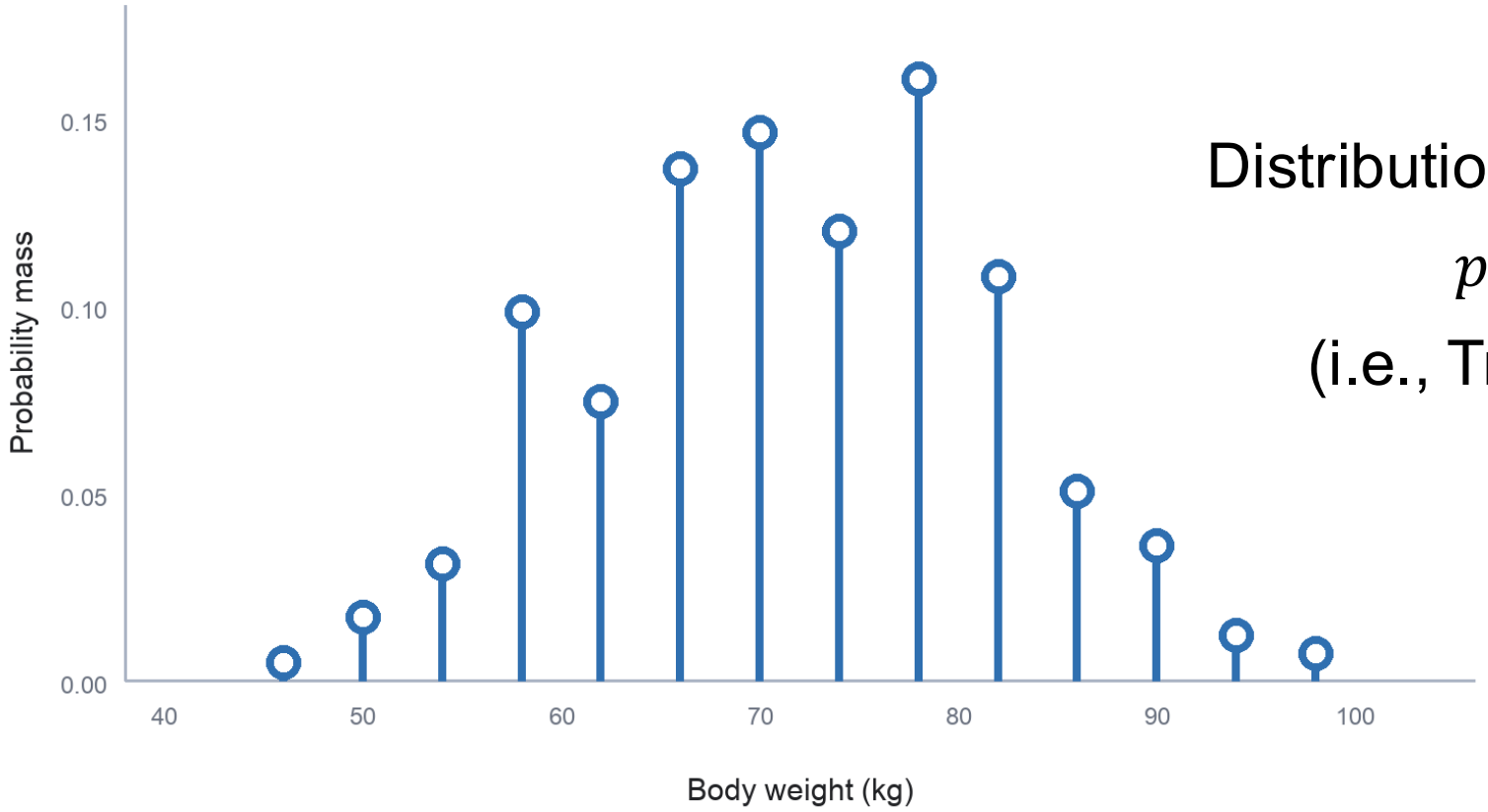
- **Question:**

How do we learn p_θ from observed samples?

(In this toy example: estimate $\boldsymbol{\theta} = [\mu, \sigma^2]^\top$)

A Toy Example: Scalar Generation

- **Question:** How do we learn p_θ?
- **Solution:** ① **Collect data samples** $x_i \sim p_{data}(x),\ i = 1,2,\dots,N$

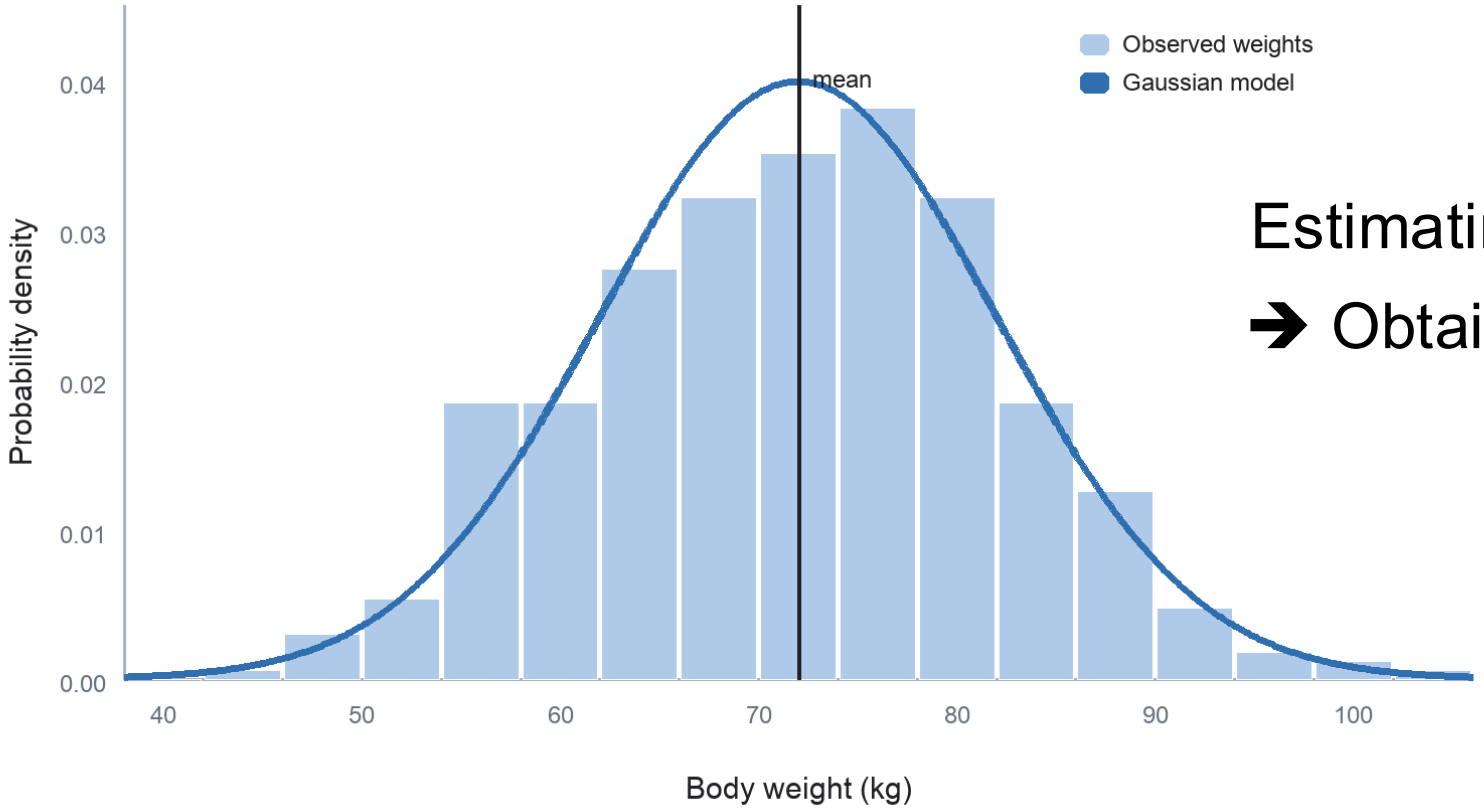


Distribution of Observed $x \sim p_{data}(x)$

(i.e., Training data)

A Toy Example: Scalar Generation

- **Question:** How do we learn p_θ?
- **Solution:** ① Collect data samples; ② **Prob. Density Estimation**

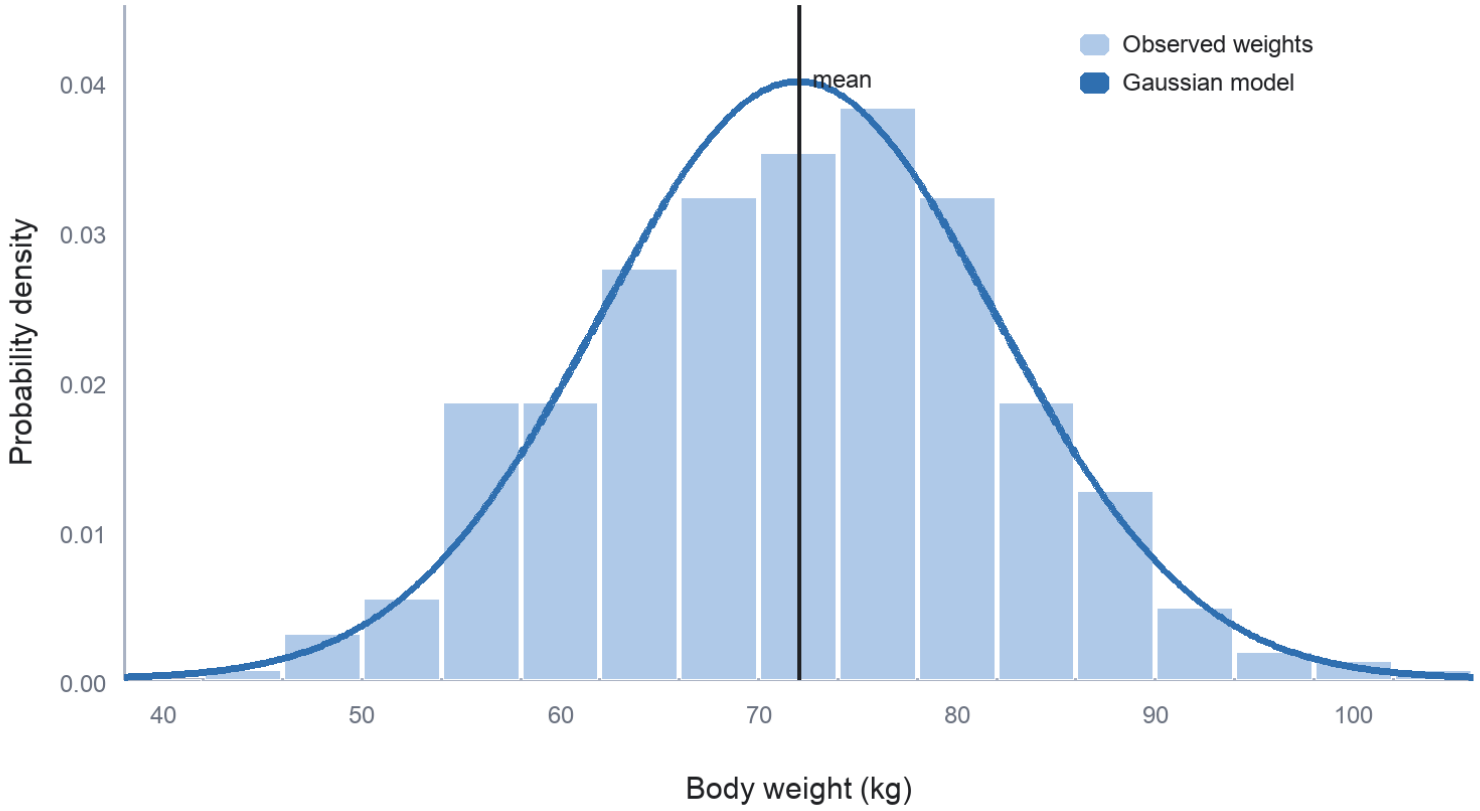


Estimating $p_\theta \approx p_{data}(x)$

➔ Obtaining $\boldsymbol{\theta} = [\mu, \sigma^2]^\top$

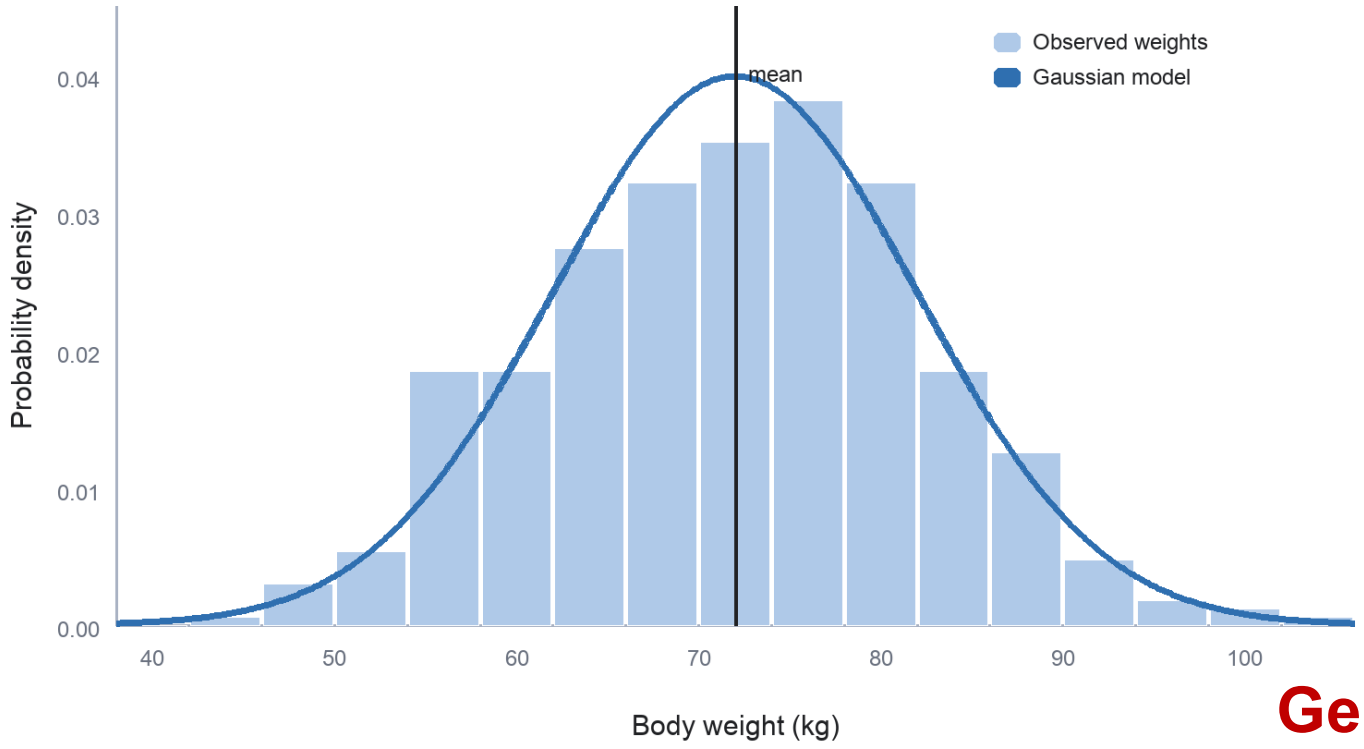
A Toy Example: Scalar Generation

- **Question:** How do we use p_θ?
- **Usage 1: Sampling** $x_{new} \sim p_\theta$ ➔ x_{new} is "plausible" as $p_\theta \approx p_{data}(x)$



A Toy Example: Scalar Generation

- **Question:** How do we use p_θ?
- **Usage 1: Sampling** $x_{new} \sim p_\theta$ ➔ x_{new} is "plausible" as $p_\theta \approx p_{data}(x)$



Generate "plausible" data

⬇

Data that are likely to be true

⬇

"Likely": Probability

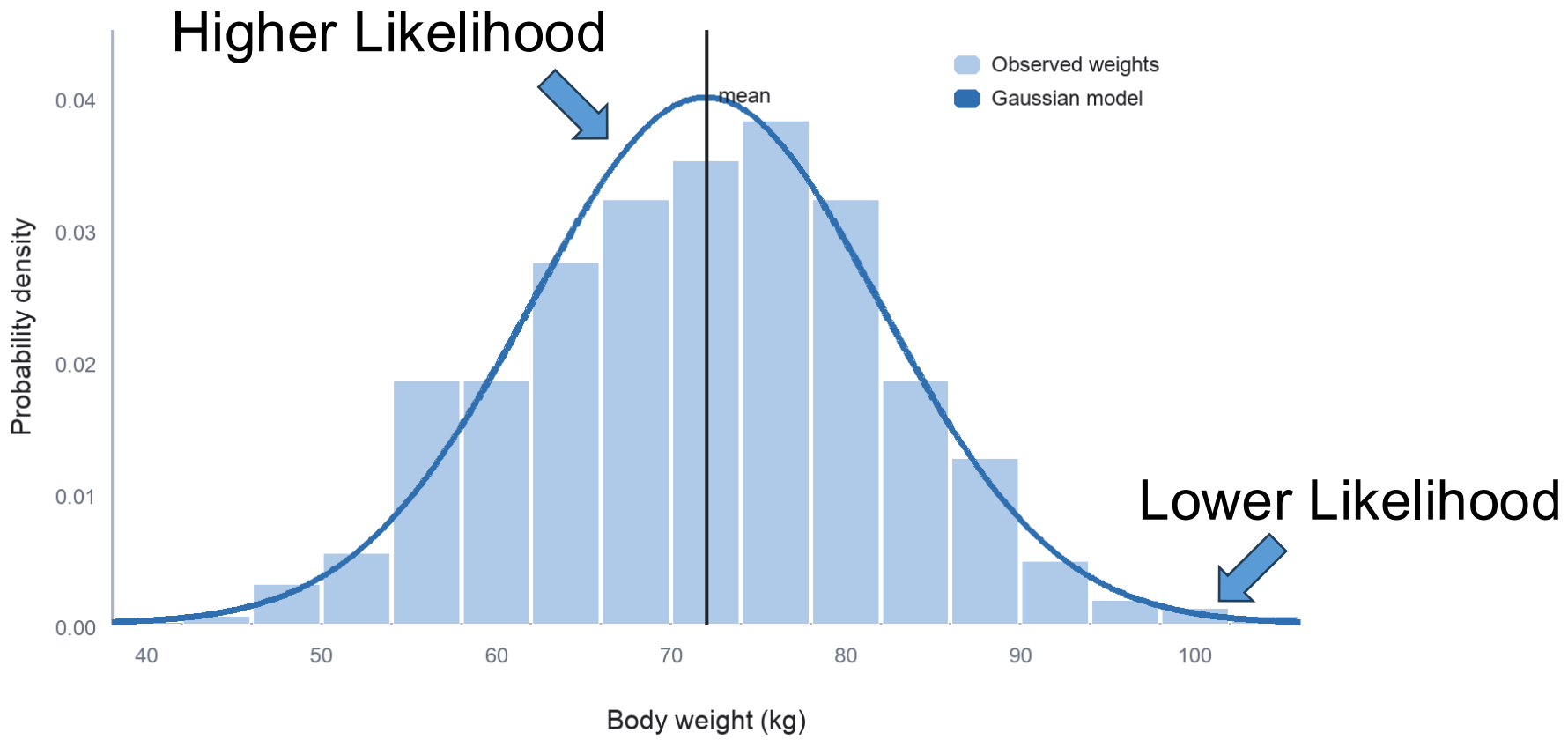
"True": Distribution p_{data}

⬇

Generative Model: Use probabilistic model to capture p_{data} by samples

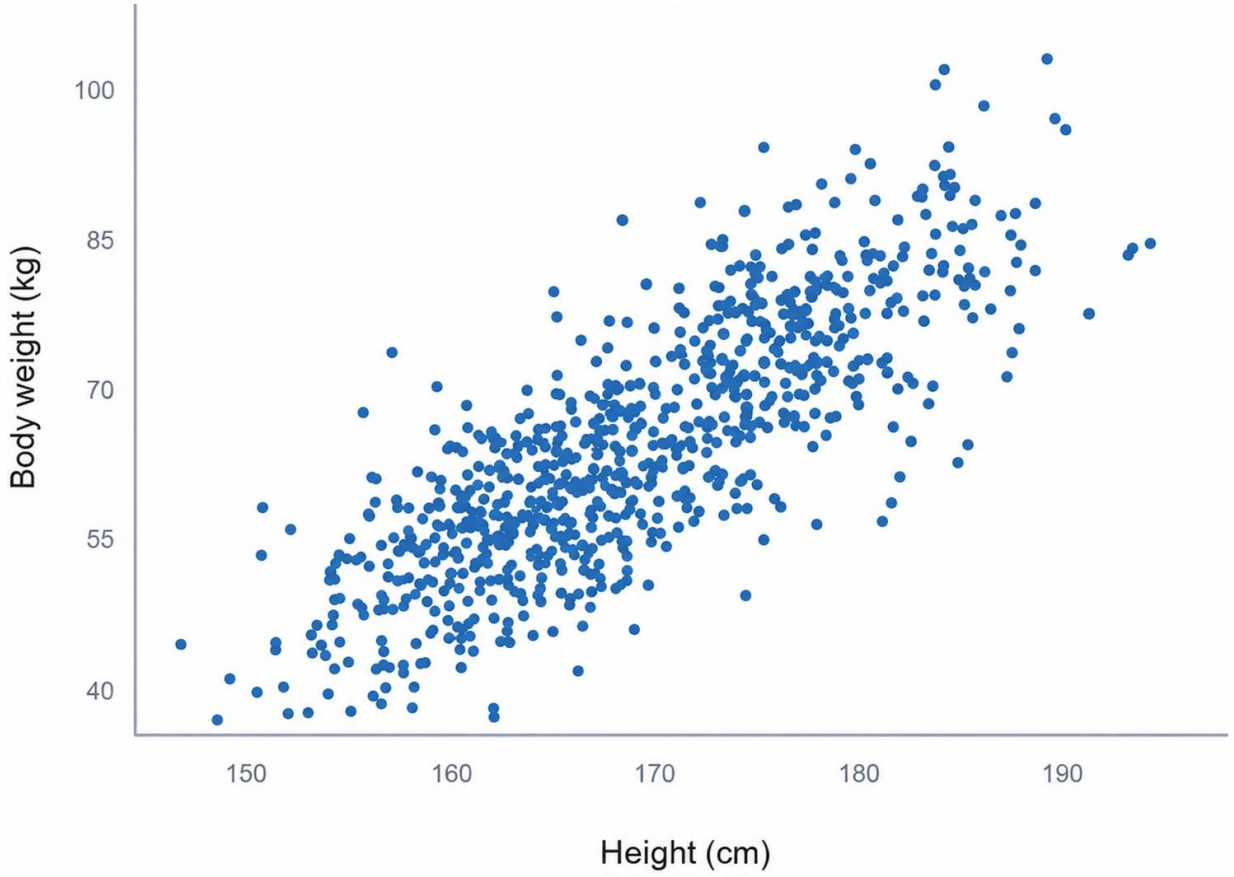
A Toy Example: Scalar Generation

- **Question:** How do we use p_θ?
- **Usage 2**: **Likelihood estimation** $p_\theta(x)$ ➔ Outlier/Anomaly Detection



A Toy Example: Scalar Generation

- **Question:** How do we use p_θ?
- **Usage 3**: **Representation Learning** $z \sim p_z$ ➔ Structural information



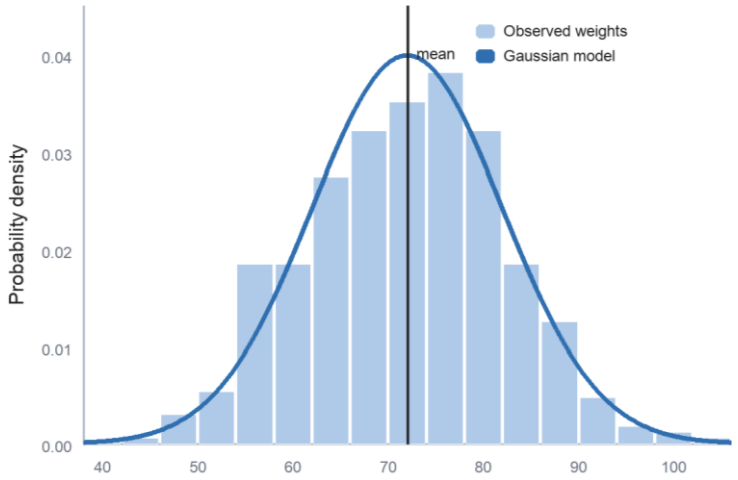
Task: Modeling <weight, height> for both male / female students

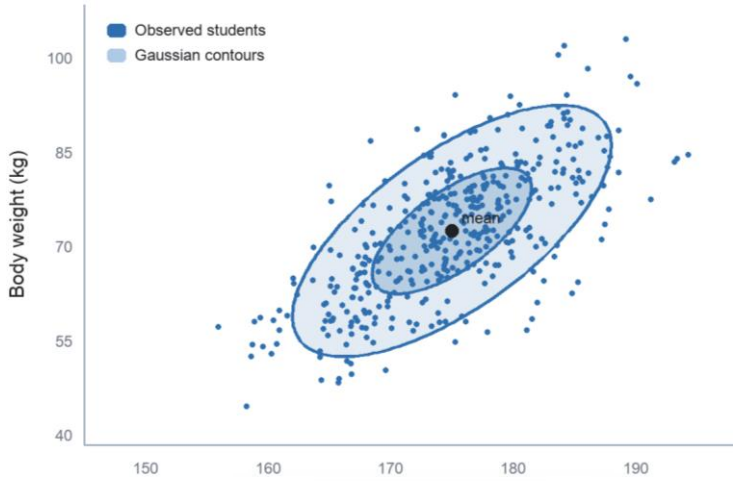
Question: **What if no label available?**

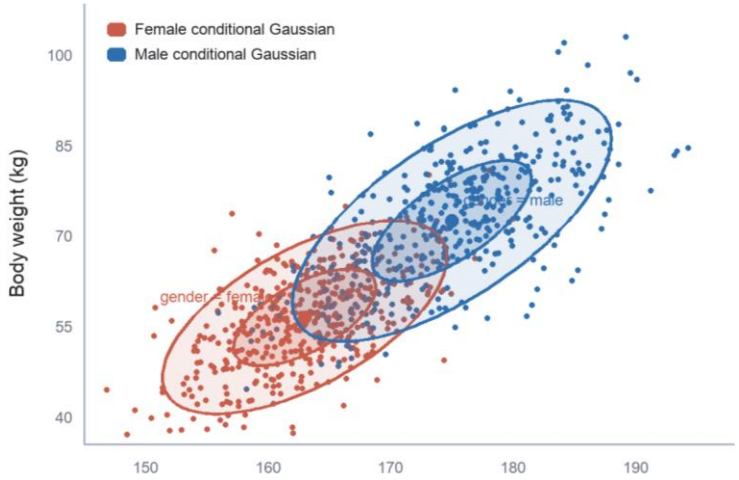
$$p_{data} = \sum_{z \in \{\mathrm{M},\mathrm{F}\}} p(x \mid z)\, p(z)$$

Latent Variable: Determines the data distribution but is **unobservable**

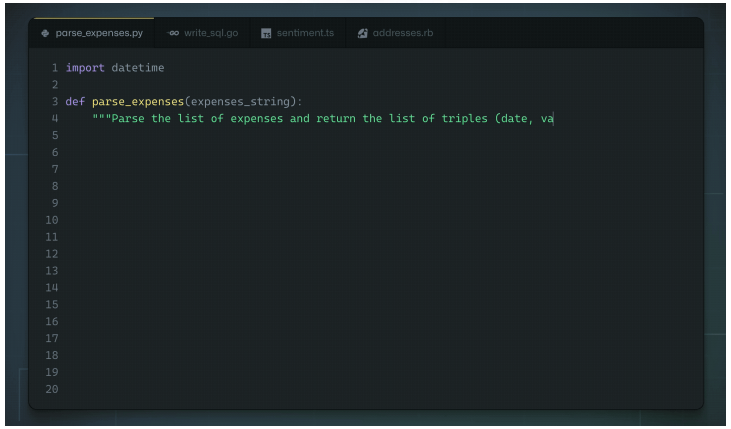
Generative Models: A Common Paradigm

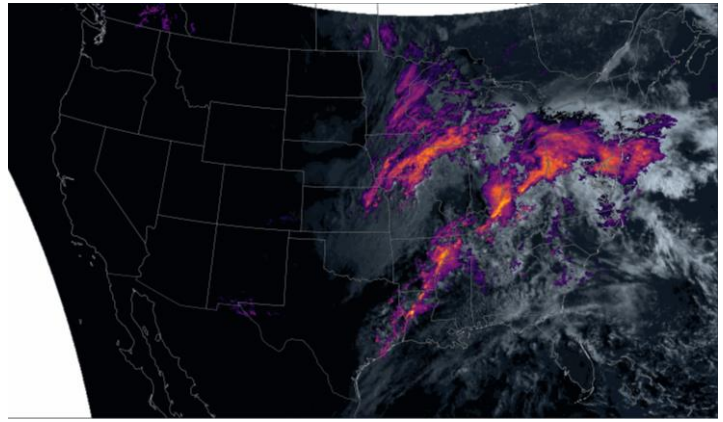












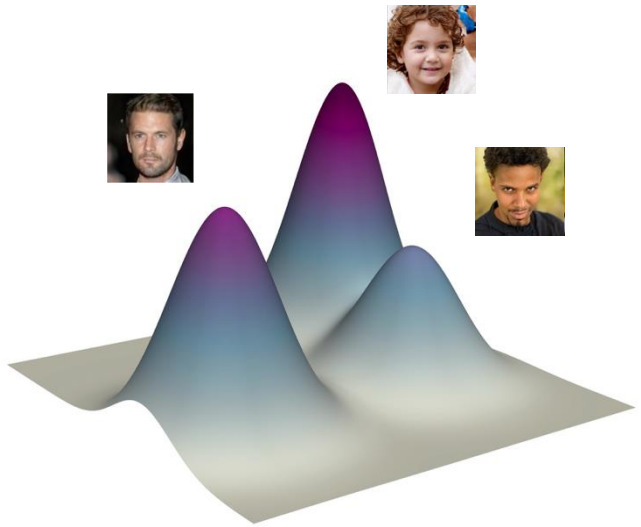
Goal: Also estimate $p_\theta \approx p_{data}(x)$, but **much more complex!**

Generative Models: A Common Paradigm

Data Samples

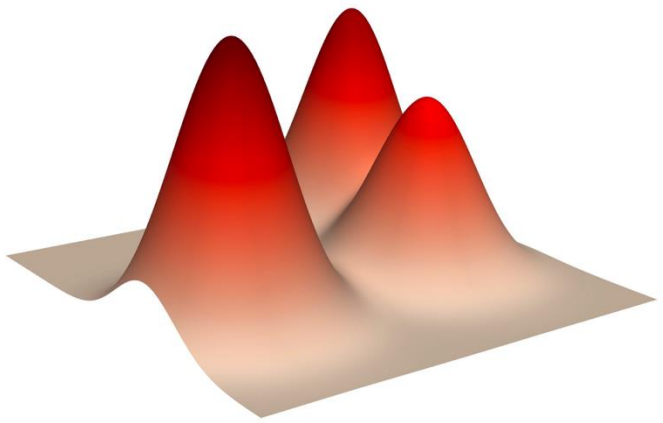
$x_i \in \mathbb{R}^d$

$i = 1,2,\ldots,N$



Data Distribution

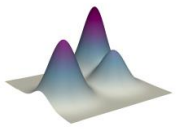
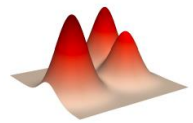
$x \sim p_{data}$



Estimated Distribution

p_θ

Optimize a loss function

$\mathcal{L}($  ,  $)$

Generative Models: A Common Paradigm

Data Samples

$\boldsymbol{x_i} \in \mathbb{R}^d$

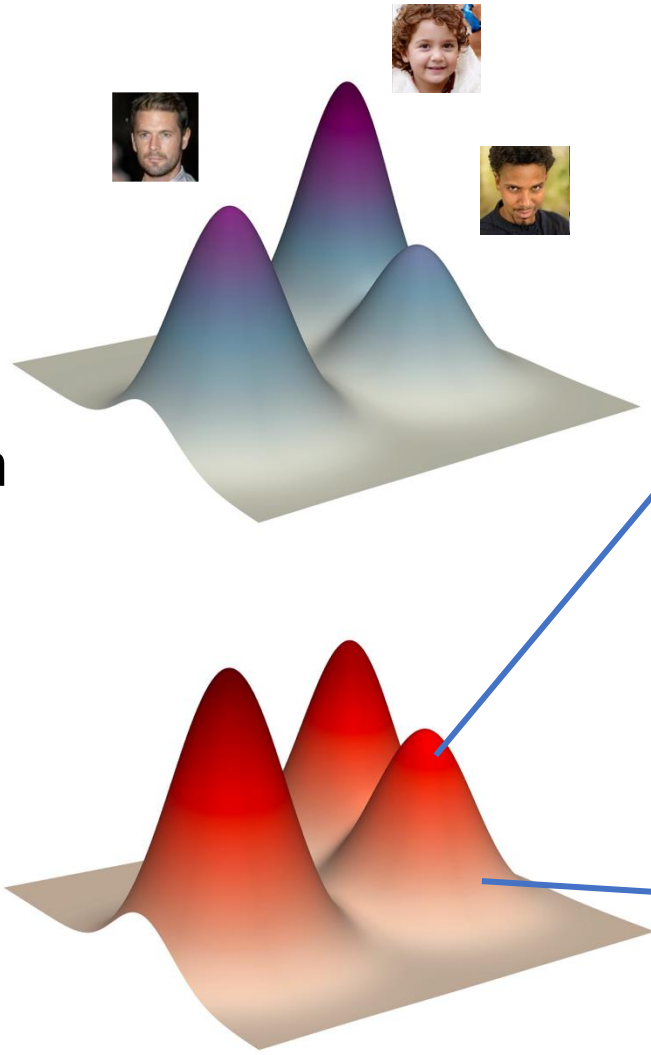
$i = 1,2, \ldots, N$

Data Distribution

$\boldsymbol{x} \sim p_{data}$

Estimated Distribution

p_θ







More plausible

(Higher likelihood)

Sample new data

$\boldsymbol{x_{new}} \sim p_\theta$

Less plausible

(Lower likelihood)

Generative Models vs. Known Systems

Generative Models vs. Discriminative Models

Data sample

$\boldsymbol{x} \sim p_{data}$

Label y

discriminative

x



model

"dog"

y

generative

y

"dog"

model



x

Label y

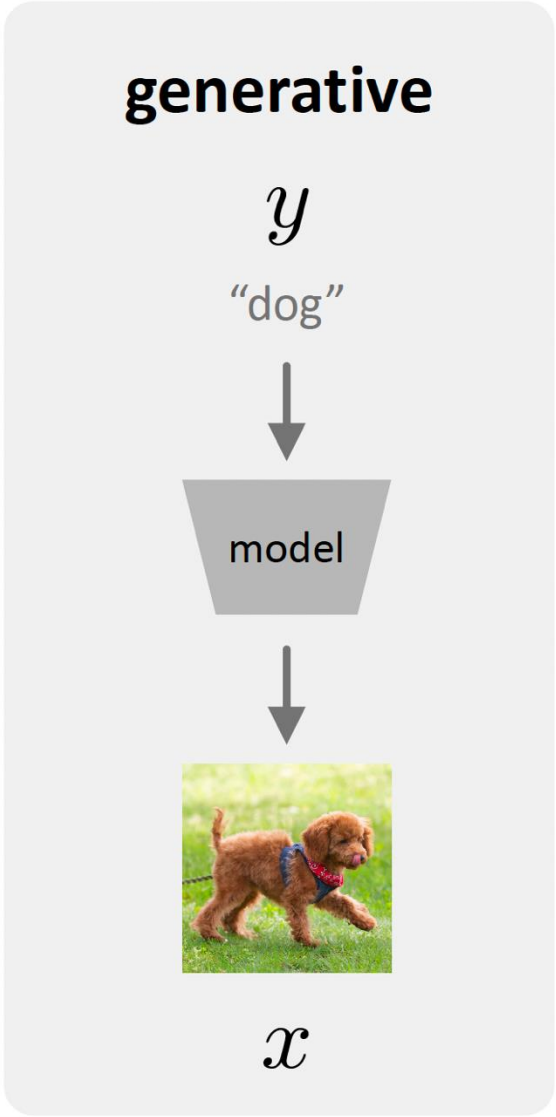
Data sample

$x \sim p_\theta$

Any difference?

Generative Models vs. Discriminative Models

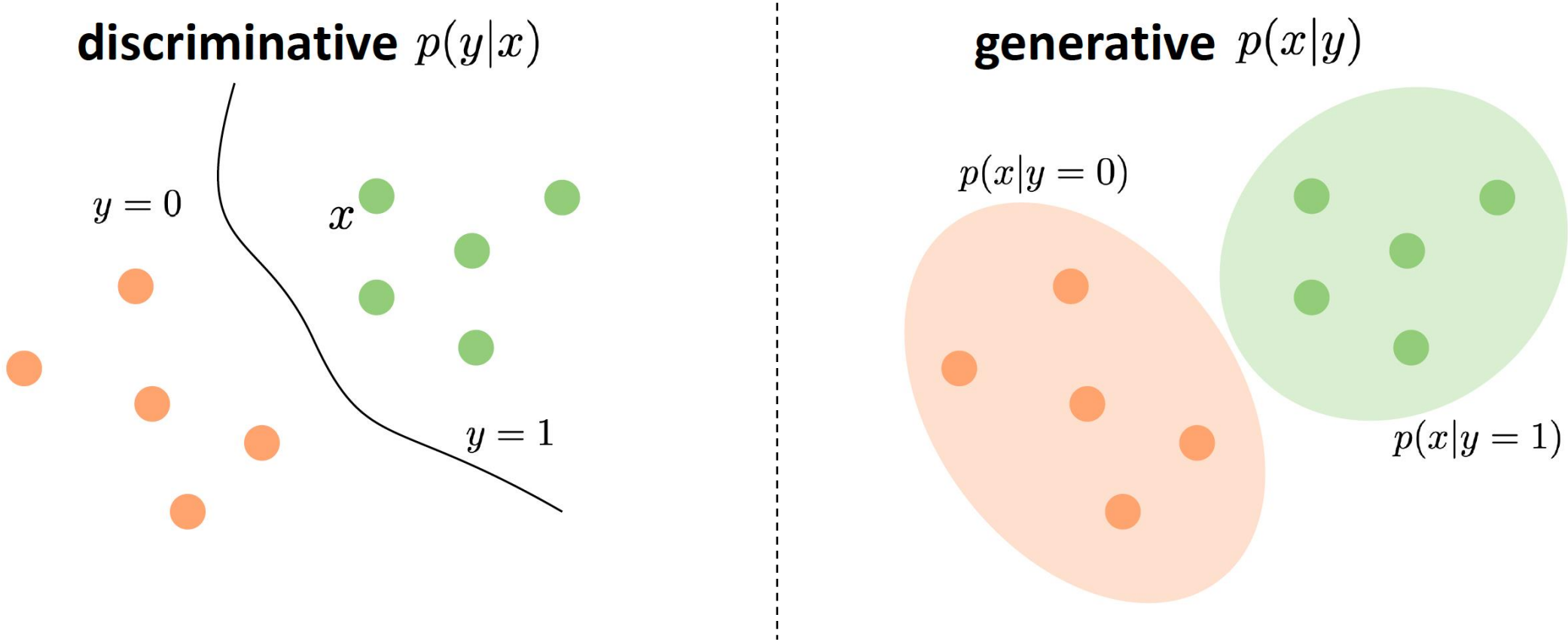
- **Property of Generative Models**
  - ➢ **Diversity:** One input y may map to **multiple** possible output x
    - ✓ Some outputs are **more plausible** than others
  - ➢ **Openness:** Training samples may contain **no ground-truth solution**
  - ➢ **Complexity:** Output x may be **higher-dimensional** than y



Label y

Data sample $x \sim p_\theta$

Generative Models vs. Discriminative Models



- Generative models apply to discriminative tasks: **Bayes' rule**
- Can discriminative models be generative?

Generative Models vs. Graphics Models

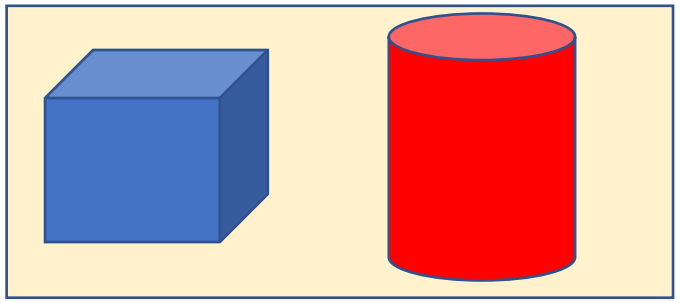
- **Task: "Generate" an image with a computer**

High level description

Cube(color=blue, position=(x,y,z), size=…)
Cylinder(color=red, position=(x',y',z'), size=..)

Generation (graphics) ⇩ ⇧ **Inference** (vision)

Raw sensory outputs



Explicit and 'rule-based' modeling (rendering)

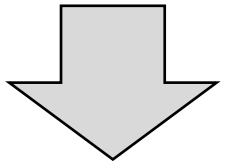
Generative Models vs. Graphics Models

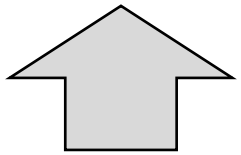
- **Task: "Generate" an image with a computer**

High level description

What parameters?

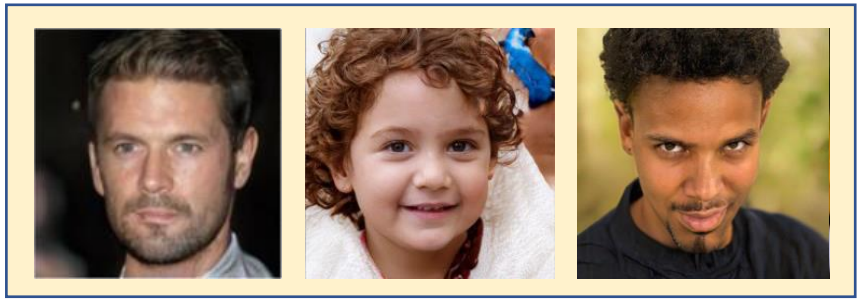
Generation (graphics)





Inference (vision)

Raw sensory outputs



- **Graphic models are generative, but are not statistical**

Generative Models vs. Graphics Models

- **Statistical** generative models are **learned from data**



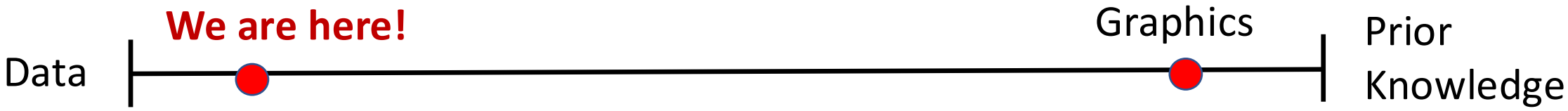
Data
(e.g., face images)

+



Prior Knowledge
(e.g., attributes, environments…)

- Priors are always necessary and helpful, but there is a spectrum



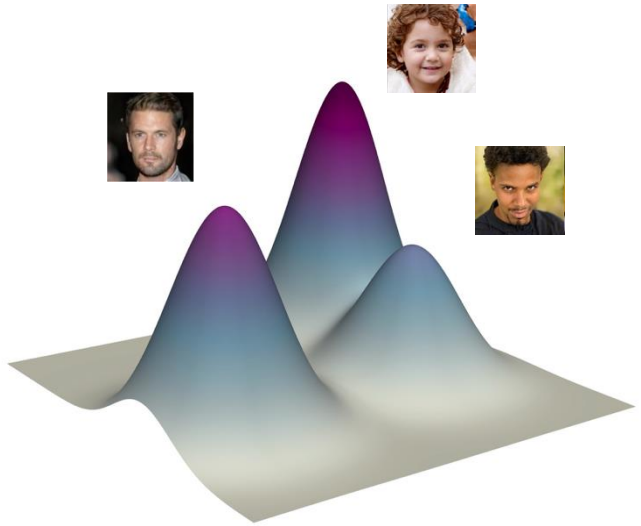
Main Challenges in Generative Models

Recall: A Common Paradigm of Generative Models

Data Samples

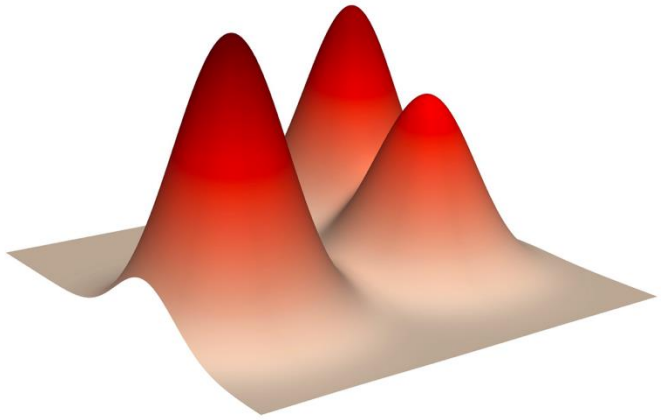
$x_i \in \mathbb{R}^d$

$i = 1,2,\ldots,N$



Data Distribution

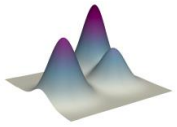
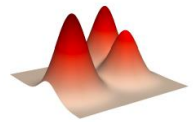
$x \sim p_{data}$



Estimated Distribution

p_θ

Optimize a loss function

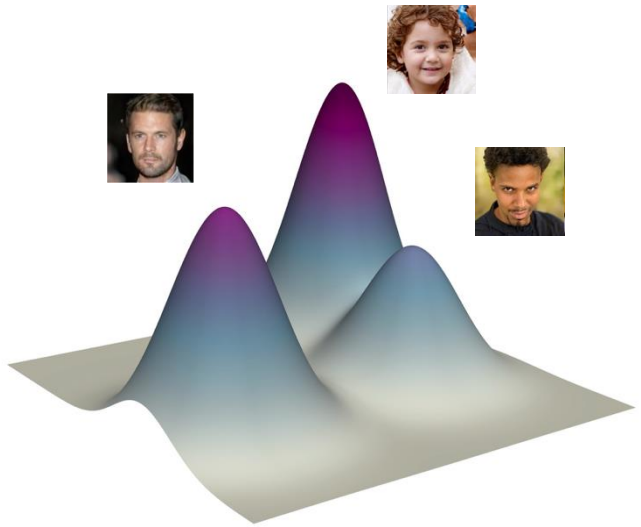
$\mathcal{L}($  ,  $)$

Main Challenges of Generative Models

Data Samples

$x_i \in \mathbb{R}^d$

$i = 1,2,\ldots,N$



Data Distribution

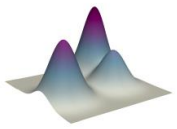
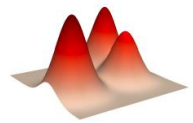
$x \sim p_{data}$

Estimated Distribution

p_θ

Challenge 1: Data dimension d can be **extremely high ($d \gg N$)**

Optimize a loss function

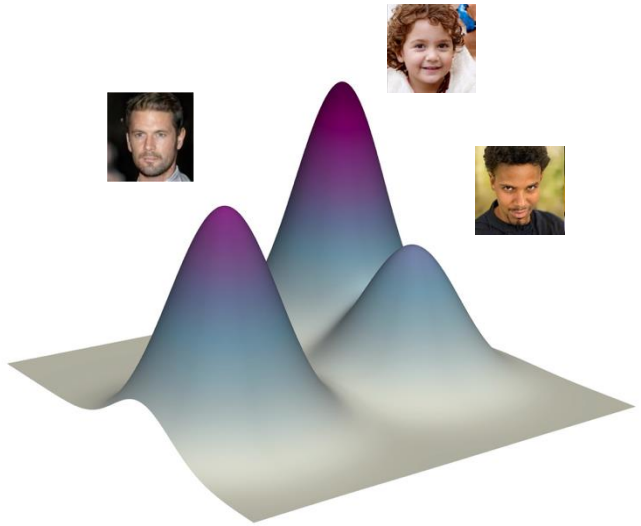
$\mathcal{L}($  ,  $)$

Main Challenges of Generative Models

Data Samples

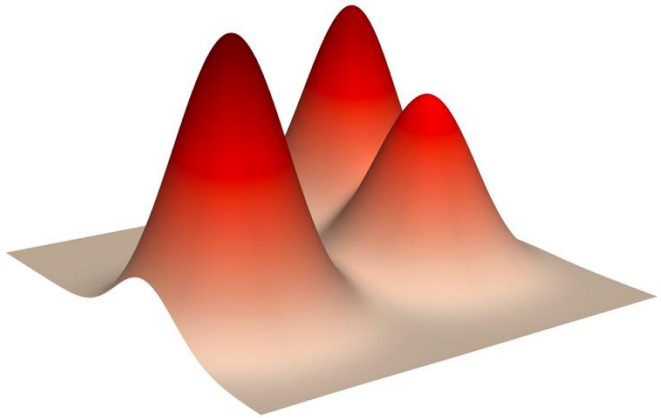
$x_i \in \mathbb{R}^d$

$i = 1,2, \dots, N$



Data Distribution

$x \sim p_{data}$

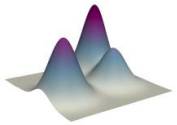
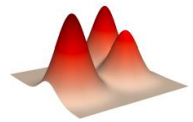


Estimated Distribution

p_θ

Challenge 2: p_{data} is **unknown**, can be **extremely complex**

Optimize a loss function

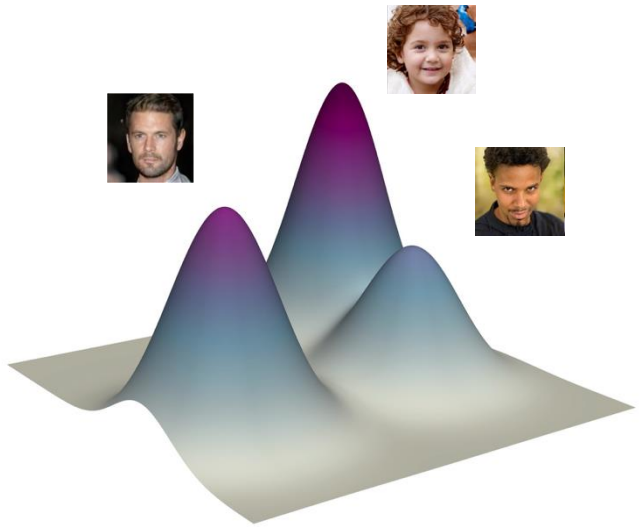
$\mathcal{L}($  ,  $)$

Main Challenges of Generative Models

Data Samples

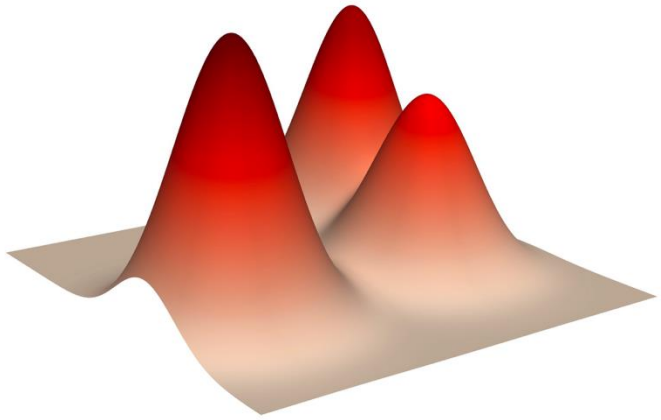
$x_i \in \mathbb{R}^d$

$i = 1,2,\ldots,N$



Data Distribution

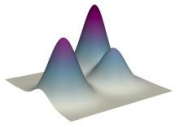
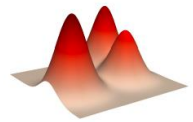
$x \sim p_{data}$



Estimated Distribution

p_θ

Optimize a loss function

$\mathcal{L}($  ,  $)$

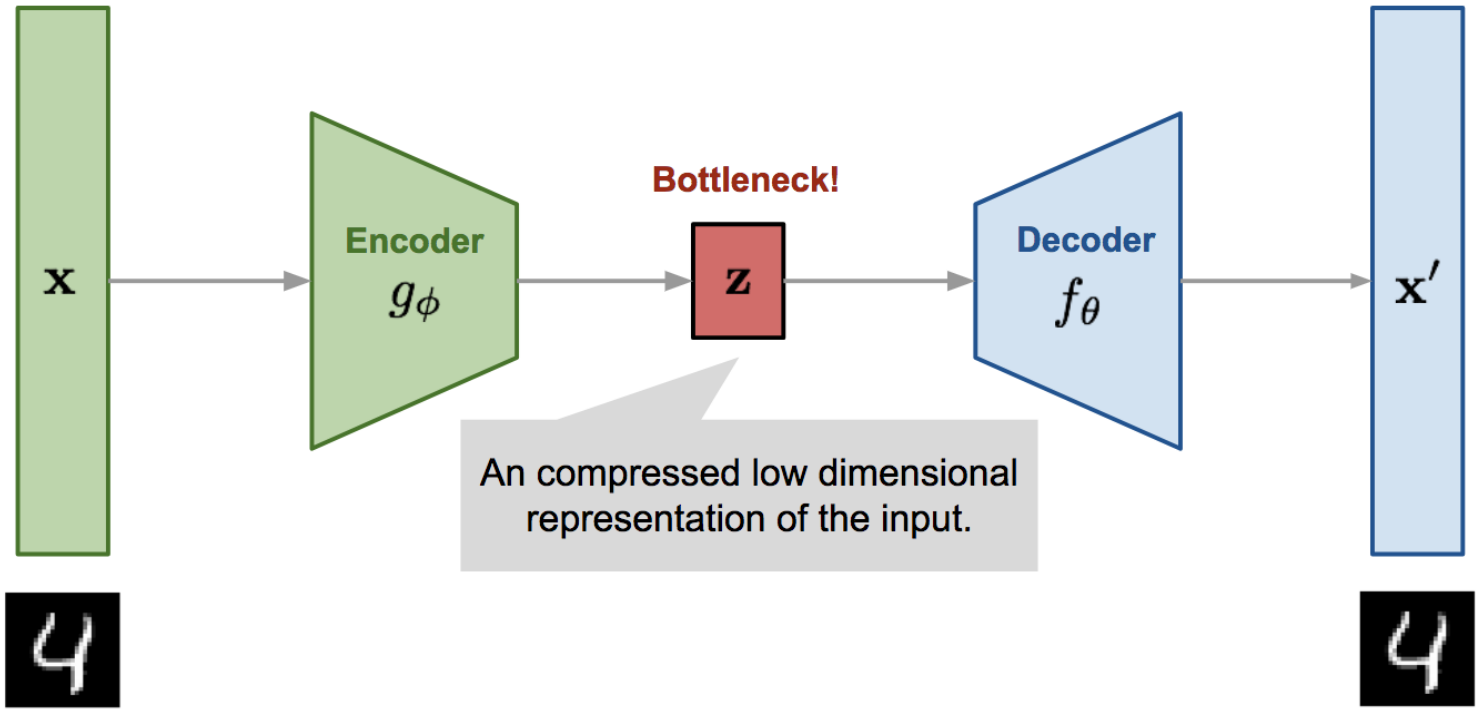
Challenge 3: Proper evaluation of the discrepancy $\mathcal{L}(p_{data}, p_\theta)$

Main Challenges of Generative Models

- **Challenge 1:** Data dimension d can be **extremely high ($d \gg N$)**

  ➔ Need **compact representations** of data samples

  ➔ Using deep neural network is one of the solutions



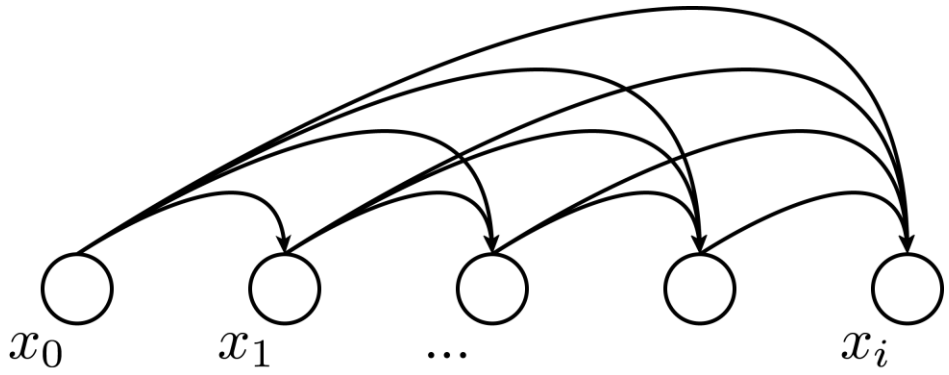
https://lilianweng.github.io/posts/2018-08-12-vae/

Main Challenges of Generative Models

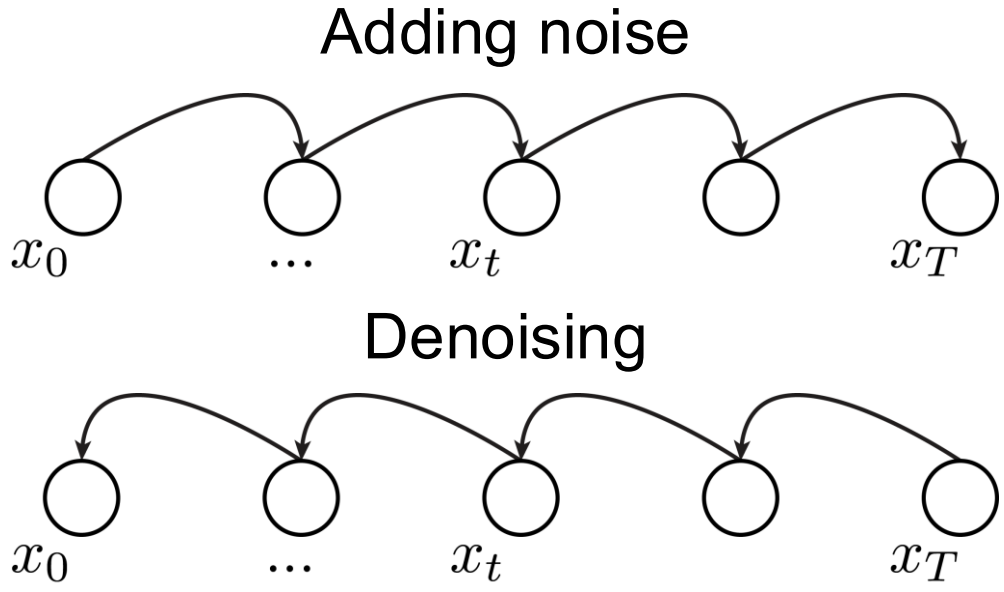
- **Challenge 2:** Data distribution p_{data} can be **extremely complex**

  ➔ **Decompose** p_{data} into simple, **tractable** components

  ➔ Learn these components with deep neural networks



Autoregressive (AR) model for sequential data



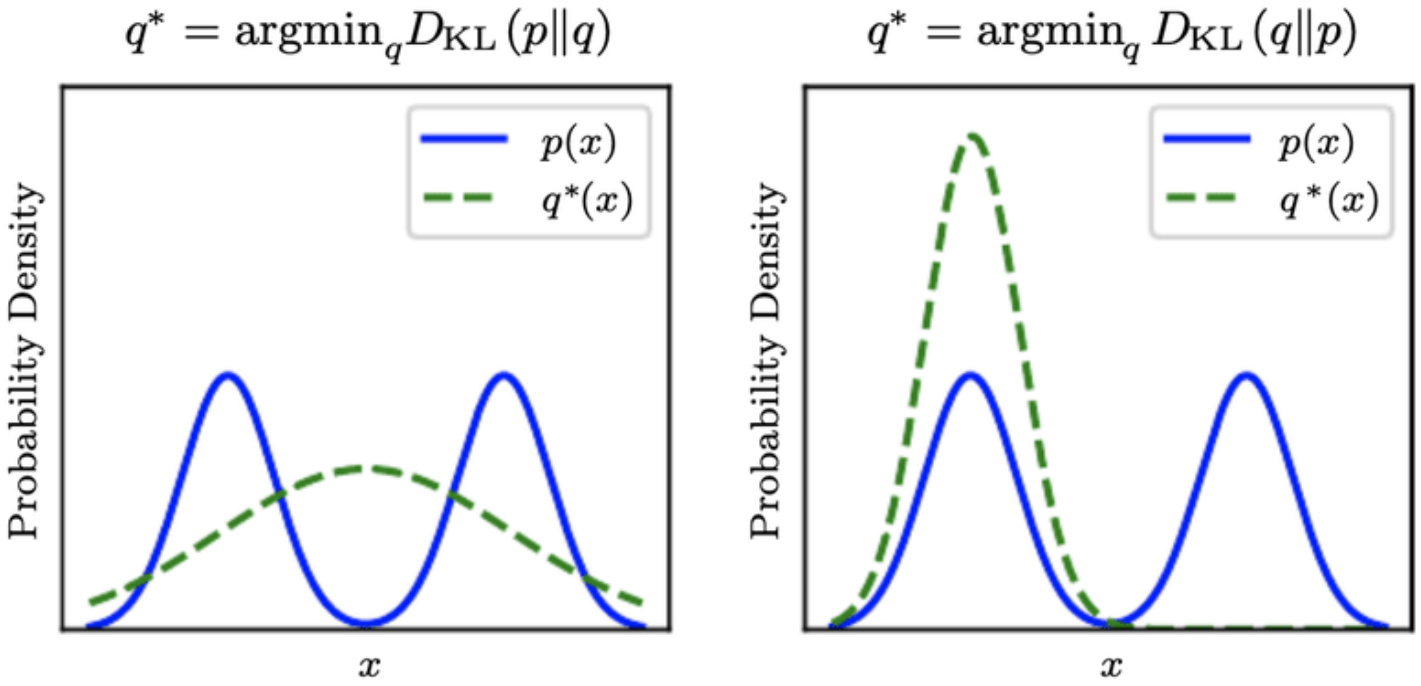
Diffusion model

Main Challenges of Generative Models

- **Challenge 3: Proper evaluation** of the gap $\mathcal{L}(p_{data}, p_\theta)$

  ➔ **No universal solution**, key to designing generative models

  ➔ Different choices lead to different model families



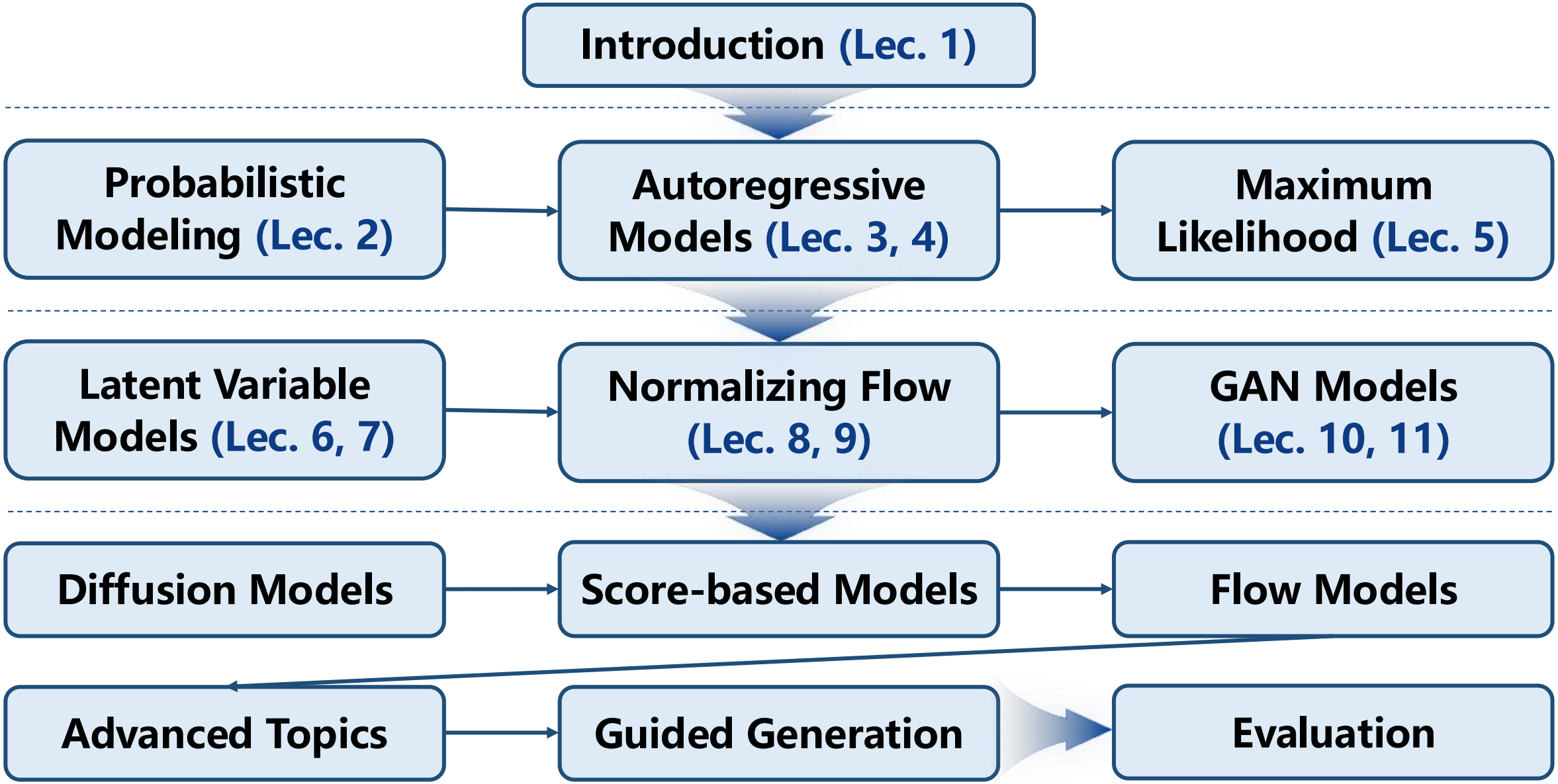
Asymmetry of KL divergence

Main Challenges of Generative Models

- **Objective:** $p_\theta \approx p_{data}$
- **Challenges:**
  - **Representation:** x is extremely **high-dimensional**
    - ➔ How to explore the **intrinsic low-dimensional structure** of $\boldsymbol{x}$?
  - **Formulation:** p_{data} is extremely **complex and has no closed form**
    - ➔ How to design p_θ that is **both expressive and tractable**?
  - **Optimization:** gap between p_θ and p_{data} is **hard to define and compute**
    - ➔ How to choose a **suitable objective** and **efficiently optimize** it?

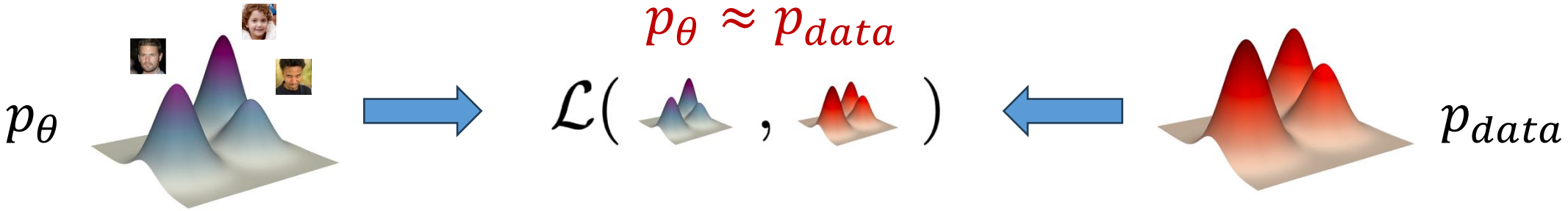
Different models address the goal and challenges differently

Roadmap


Introduction (Lec. 1)
Probabilistic Modeling (Lec. 2)
Autoregressive Models (Lec. 3, 4)
Maximum Likelihood (Lec. 5)
Latent Variable Models (Lec. 6, 7)
Normalizing Flow (Lec. 8, 9)
GAN Models (Lec. 10, 11)
Diffusion Models
Score-based Models
Flow Models
Advanced Topics
Guided Generation
Evaluation

Summary

- **Generative models are statistical and probabilistic**
- **Objective**



- **Usage:**
  - Sampling
  - Likelihood Estimation
  - Representation Learning
- **Challenges:**
  - Representation
  - Formulation
  - Optimization

Supplementary Material

- **Stanford CS 236 Lecture 1**
  - https://www.youtube.com/watch?v=XZ0PMRWXBEU&list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8
- **Cornell CS 6785 Lecture 1**
  - https://www.youtube.com/watch?v=lZgvgLy1wyg&list=PL2UML_KCiC0UPzjW9BjO-lW6dqliu9O4B
- **Deep Generative Models**
  - **Section 1**
- **The Principles of Diffusion Models**
  - **Section A.1**

Reading List

- **Reading List:**
  - D. Zoran and Y. Weiss, From Learning Models of Natural Image Patches to Whole Image Restoration, ICCV 2011
  - D. Zoran and Y. Weiss, Natural Images, Gaussian Mixtures and Dead Leaves, NeurIPS 2012
  - D. Ulyanov et al., Deep Image Prior, CVPR 2018
- **(Optional) Recommended readings:**
  - A. A. Efros and T. K. Leung, Texture Synthesis by Non-parametric Sampling, 1999
  - C. Barnes et al., PatchMatch: A Randomized Correspondence Algorithm for Structural Image Editing, 2009
  - W. T. Freeman et al., Example-Based Super-Resolution, 2002